

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	2	3	2	3	1	3	2	1	1	-	2	1	2	3	2
CO2	1	3	3	3	3	2	3	3	3	2	-	3	1	2	3	1
CO3	1	2	3	3	3	2	3	3	3	3	-	3	1	2	3	1
CO4	1	2	3	2	3	3	1	2	1	2	-	3	1	2	2	2
CO5																

Teaching-Learning Strategies in brief (4-5 sentences):

The course will combine lectures with a range of interactive and applied learning activities to strengthen both conceptual understanding and empirical skills. Lectures will introduce key theoretical frameworks, debates, and empirical findings on the relationship between social identity and economic development. Students will engage critically with assigned readings through structured reading responses and classroom discussions, examining how social identity shape economic opportunities and outcomes. The course will also incorporate applied data analysis exercises that enable students to investigate patterns of inequality and engage directly with empirical evidence. Together, these components are designed to foster critical thinking, and the ability to connect theoretical perspectives with real-world processes of inequality, inclusion, and economic change.

Title of the Course : Information Retrieval and Extraction

Name of the Faculty : Anil Nelakanti

Course Code : CS4.406

L-T-P : 3-1-1

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program: B.Tech. in Computer Science and Engineering

1.Prerequisite Course / Knowledge:

Basic principles of Computer programming, Statistical Methods in Artificial Intelligence, Programming experience in Python, and Algorithms.

2.Course Outcomes (COs)

After completion of this course successfully, the students will be able to...

CO-1. Develop algorithms to retrieve information from unstructured data

CO-2. Design and architect information retrieval systems for worldwide web

CO-3: Design Web crawling systems

CO-4. Design algorithms to process noisy data in document repositories

CO-5. Develop information extraction system's

3.Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	1	1	-	2	2	2	2	3	3	2	2
CO2	3	3	3	3	3	1	1	1	2	2	2	2	3	3	2	2
CO3	3	3	3	3	2	1	1	1	2	2	2	2	3	3	2	2
CO4	3	3	3	3	2	1	1	-	3	2	2	2	3	3	2	2
CO5	3	3	3	3	2	1	1	-	3	2	2	2	3	3	2	2

‘3’ in the box denotes ‘High-level’ mapping, 2 for ‘Medium-level’ mapping, 1 for ‘Low’-level’ mapping

4.Detailed Syllabus:

Unit 1: Introduction to Information retrieval, Information Extraction and Information Access systems. (6 hours)

Unit 2: Information Retrieval Models and Evaluation of IR systems (7.5 hours);

Unit 3: Web Information Retrieval (4.5 hours)

Unit 4 Natural Language Processing in IR (7.5 hours)

Unit 5: Machine Learning in Information Retrieval Systems (12 hours)

Unit 6: Information Extraction (4.5 Hours)

Unit 7: IR Applications (12 Hours)

References:

1. Introduction Information Retrieval – Chris Manning et al (the Stanford IR Book) (ISBN-13: 978-0521865715)
2. Search Engines: IR in Practice – Bruce Craft et al (ISBN-13: 978-0136072249)
3. Research papers

5.Teaching-Learning Strategies in brief:

Lectures by integrating ICT into classroom teaching, weekly tutorials involving problem solving and active learning by students and Project-based Learning by doing one mini-project and a major project by the students.

6.Assessment methods and weightages in brief:

Assignments (theory and programming): 20 marks

Seminar: 10 marks

Project: 35 marks

Mid Semester Examination: 15 marks

End Semester Examination: 20 marks

Karma Points: Exceptional performance, active participation in class/tutorials, and help provided to other students, which a TA or instructor notices deem worthy of 3 marks.

Title of the Course	: Information, Computation, and Learning
Faculty Name	: Prasad Krishnan
Name of the Program	: ECE (Also intended as Open Elective for ECE/CSE)
Course Code	: EC5.415
Credits	: 3:1:0:4
L - T - P	:
(L - Lecture hours, T-Tutorial hours, P - Practical hours)	

Semester, Year : **Monsoon 2026**

Pre-Requisites : Basics of Probability (Discrete and Continuous Random Variables, pdfs/pmfs, Gaussian. Course is meant to be accessible to new MS/PhD students also, apart from B.Tech/DD in CSE/ECE.

Prior courses such as Mathematics of Information (UG2-half), Information and Communication (UG1-ECE) can be useful but not mandatory.

Note: Probability review, Discrete Entropy and its Properties – elaboration on these initial topics will be moved to tutorial sessions for those students who have not seen them in a course before.

Course Outcomes :

1. Explain various fundamental quantities in Information-theory and analyze their mathematical properties and utility.
2. Formulate Data Compression and Communication through Channels as Information-theory-based problems and characterize their fundamental limits. .
3. Apply information theoretic ideas to study the fundamental limits of secure computation and various related sub-areas.
4. Apply information-theoretic ideas for understanding machine-learning and improving the same.
5. Comprehend, write, and communicate academic research in Information-theory and applications, effectively as an individual or groups, while executing a project or a research paper.

Course Topics:

Unit 1: Basics of Information Theory : Entropy and Mutual Information, KL Divergence and its properties, Differential Entropy and its properties, Gaussian Entropy as the largest entropy for any RV with fixed variance,